

Department: **Data Science and Artificial Intelligence**
Course Name: **Data Science Tools and Software**
Course Code: **03203N**
Professor Name: **Dr. Laila M. Fatehy**
Level: **Two**



Sheet 5 (Data Cleaning)

1) Giving the following Data Frame of Markets

	Meat	milk	egg	orange
market 1	100	20	10	NaN
market 2	120	NaN	12	15
market 3	150	15	NaN	10

a) write a simple python code to Create a new DataFrame named market_df

b) count the number of non-NaN (non-Missing) values

c) use a method that Return boolean True/False for each element if it is a NaN

d) count the number of Missing values in each column

e) Calculate the total number of Missing values in DataFrame

f) Delete rows having Missing values

g) Delete Columns having NaN values

h) Replace each NaN value with 0

i) Solve Missing Value Problem by Using Forward fill method

j) Solve Missing Value Problem by Using Backward fill method

2) Use (simple_data.csv) Dataset and write a python code for the following:

a) Describe your Dataset

b) Replace missing values in “price” column by column mean

c) Replace NaN in “Age” column by column median

d) use a method that Return boolean T or F for each row if exist duplicates of rows

e) Calculate the total number of duplicating rows in “simple_data”

f) Clean your dataset from duplicating rows

Good Luck
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